# Extended Case 4 — Perfect Parachute

Compare four suppliers with ANOVA · 307312 Applied Statistics — Masters Level

- **Start after:** independent-group comparisons and more than two means in Part 3.
- **Suggested time:** 70–90 minutes, including the ANOVA introduction below.
- **Business question:** Do the four suppliers differ in mean tensile strength, and what evidence should guide procurement?

**Learning outcomes:**

- Compare variation within and between groups.
- Interpret an overall ANOVA test and adjusted pairwise comparisons.
- State assumptions and avoid equating statistical significance with supplier suitability.

## 1. The procurement manager's brief

- Perfect Parachute purchases synthetic fibre from four suppliers.
- Five parachutes per supplier are woven and tested to breaking point.
- **Source:** *Basic Business Statistics*, ninth edition, Chapter 11, pp. 388 and 393–399.
- **Unit:** one tested parachute; 20 measurements in four groups.
- **Outcome:** tensile strength on the source scale; a physical force unit is not specified in the reviewed material.
- **Limit:** small groups provide limited information about distribution shape and supplier reliability.

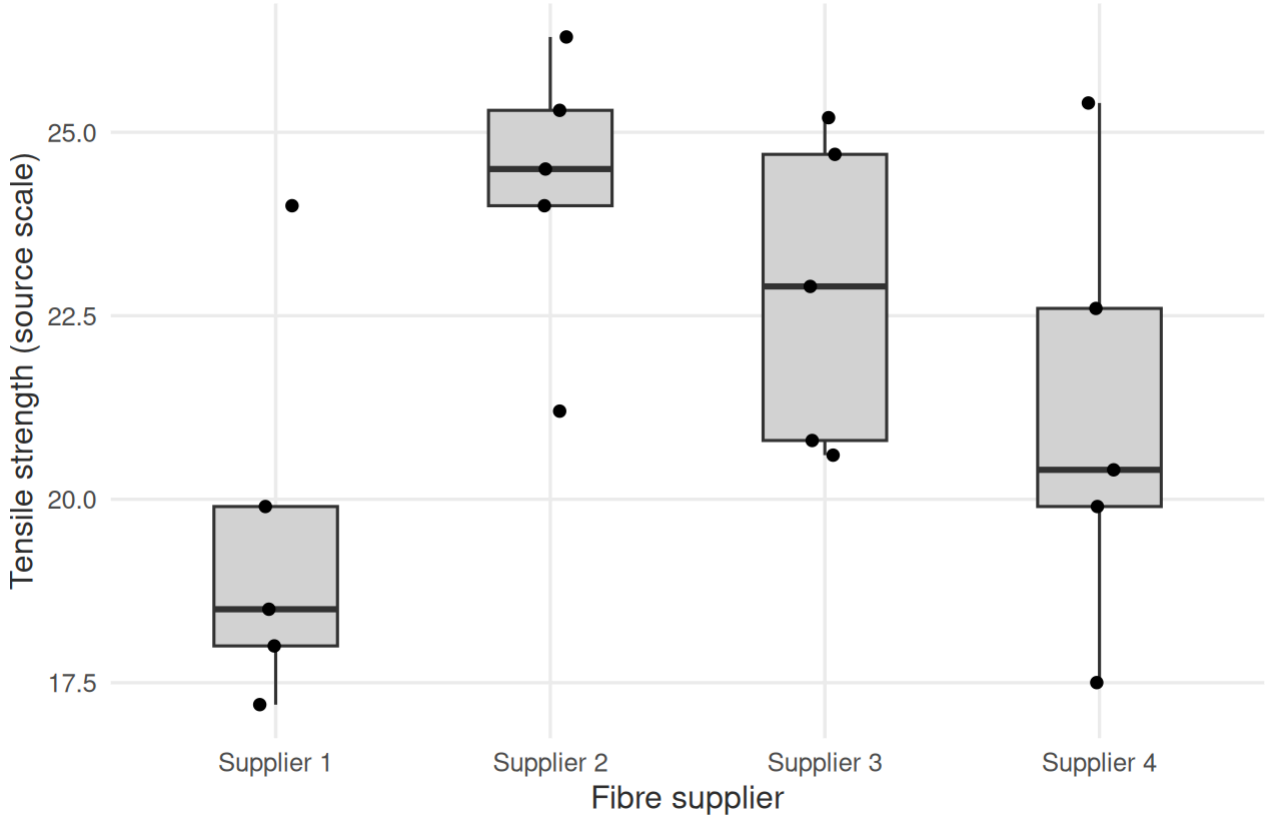
## 2. Import measurement rows only

```
data_dir <- if (dir.exists("data")) "data" else file.path("..", "data")
source(file.path(data_dir, "..", "lessons", "_includes", "course-tools.R"))
parachutes <- read.csv(file.path(data_dir, "business", "extended-parachute.csv"))
parachutes$supplier <- factor(parachutes$supplier)
stopifnot(nrow(parachutes) == 20, all(table(parachutes$supplier) == 5))
supplier_summary <- parachutes %>% group_by(supplier) %>%
  summarise(n = n(), mean_strength = mean(strength), sd_strength = sd(strength),
    min_strength = min(strength), max_strength = max(strength), .groups = "drop")
knitr::kable(supplier_summary, digits = 2)
```

| supplier | n | mean_strength | sd_strength | min_strength | max_strength |
|---|---|---|---|---|---|
| Supplier 1 | 5 | 19.52 | 2.69 | 17.2 | 24.0 |
| Supplier 2 | 5 | 24.26 | 1.92 | 21.2 | 26.3 |
| Supplier 3 | 5 | 22.84 | 2.13 | 20.6 | 25.2 |
| Supplier 4 | 5 | 21.16 | 2.98 | 17.5 | 25.4 |

- `data/business/extended-parachute.csv`; guide: `data/business/README.md`.
- Workbook means, medians and standard deviations are summary rows; they are excluded from the 20 observations.
- The means are 19.52, 24.26, 22.84 and 21.16 for suppliers 1–4.

```
ggplot(parachutes, aes(supplier, strength)) +
  geom_boxplot(fill = petra_accent, width = .45, outlier.shape = NA) +
  geom_point(position = position_jitter(width = .06, height = 0, seed = 307312)) +
  labs(x = "Fibre supplier", y = "Tensile strength (source scale)")
```




- Individual points keep the five observations per supplier visible.
- A higher sample mean does not by itself establish a superior supplier.

## 3. A short ANOVA introduction

- Separate variation **between supplier means** from variation **within suppliers**.
- ANOVA divides each sum of squares by its degrees of freedom to form a mean square.
- **F = between-group mean square / within-group mean square.**
- A large F is evidence against equal population means under the model.
- **Null:** all four population mean strengths are equal.
- **Alternative:** at least one differs; the test does not say that all pairs differ.
- Classical one-way ANOVA assumes independent errors, approximately normal errors within groups and a common error variance.

```
fit <- aov(strength ~ supplier, data = parachutes)
summary(fit)
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
supplier     3  63.29  21.095   3.462 0.0414 *
Residuals   16  97.50   6.094
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- `aov()` fits the one-factor model; supplier is categorical, not a numeric trend.

- F is approximately **3.462**, with p approximately **0.0414**.
- At 5%, the model gives evidence against all four means being equal.

## 4. Which comparisons are supported?

```
TukeyHSD(fit)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

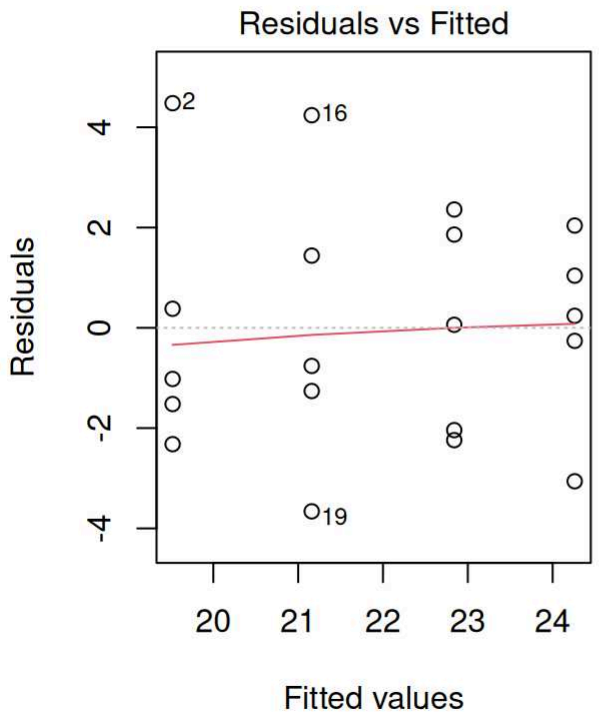
Fit: aov(formula = strength ~ supplier, data = parachutes)

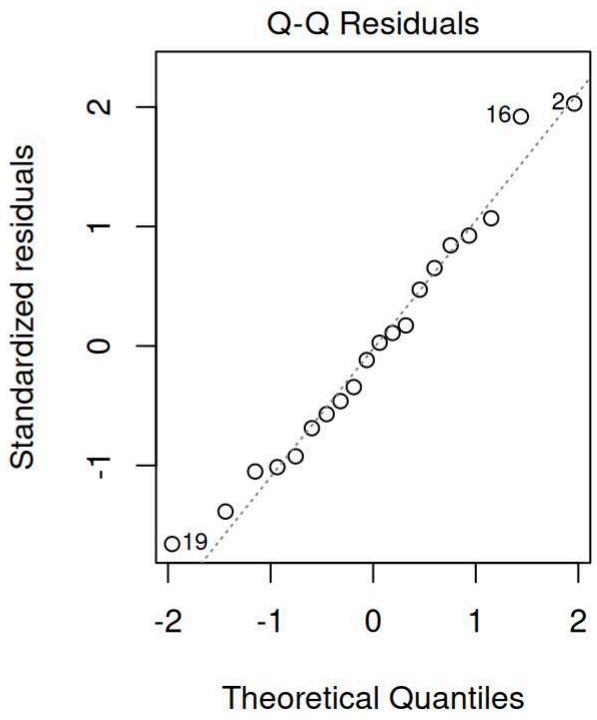
$supplier
                       diff        lwr      upr     p adj
Supplier 2-Supplier 1  4.74  0.2731426 9.206857 0.0356281
Supplier 3-Supplier 1  3.32 -1.1468574 7.786857 0.1868706
Supplier 4-Supplier 1  1.64 -2.8268574 6.106857 0.7231935
Supplier 3-Supplier 2 -1.42 -5.8868574 3.046857 0.8000982
Supplier 4-Supplier 2 -3.10 -7.5668574 1.366857 0.2340709
Supplier 4-Supplier 3 -1.68 -6.1468574 2.786857 0.7084910
```

- Tukey's procedure adjusts pairwise intervals and tests for the family of comparisons under the ANOVA assumptions.
- Read each contrast's direction and interval; do not rank suppliers using p-values alone.
- An overall significant test need not make every pair significant.
- These intervals concern mean differences, not minimum guaranteed product strength.

## 5. Check the model and its limits

```
par(mfrow = c(1, 2))
plot(fit, which = 1)
plot(fit, which = 2)
```




```
par(mfrow = c(1, 1))
fligner.test(strength ~ supplier, data = parachutes)
```

```
    Fligner-Killeen test of homogeneity of variances

data:  strength by supplier
Fligner-Killeen:med chi-squared = 0.55853, df = 3, p-value = 0.9059
```

- Residual plots assess spread and approximate normality; they cannot establish independent assignment.
- `fligner.test()` checks dispersion; a large p-value does not prove equal variances.
- Five observations per supplier give imprecise diagnostics.
- Causal supplier claims need a defensible assignment protocol and control of weaving/testing conditions.

## Optional: relax the equal-variance assumption

```
oneway.test(strength ~ supplier, data = parachutes, var.equal = FALSE)
```

```
    One-way analysis of means (not assuming equal variances)

data:  strength and supplier
F = 3.3206, num df = 3.0000, denom df = 8.7743, p-value = 0.07206
```

- Welch's one-way test avoids assuming equal variances.
- Tukey intervals from the classical model do not automatically become Welch-compatible pairwise intervals.
- Report sensitivity openly rather than choosing the method giving the preferred answer.

## 6. Independent procurement briefing

1. Explain F using the observed between- and within-supplier variation.
2. Identify the largest sample mean difference and report its Tukey interval and adjusted p-value.
3. Compare classical and Welch conclusions, including their assumptions.
4. Explain why five tests per supplier cannot establish rare failure rates or minimum guaranteed strength.
5. Specify additional evidence on cost, delivery reliability and engineering requirements before recommending a supplier.
6. Propose a follow-up experiment controlling loom conditions and assigning fibre treatments appropriately.

## 7. Optional design discussion: supplier and loom

- The book also considers four suppliers and two loom types.
- The local `Parachute2` workbook contains a separate `Data` sheet with loom-level observations as well as summary sheets.
- This case analyses the original **one-way, 20-measurement dataset**; it does not combine the two experiments.
- A supplier-by-loom interaction means the supplier difference depends on loom type.
- Discuss how an interaction would change procurement advice; a two-factor analysis is beyond this assignment.

## Submission

- Supplier summary table, one chart and a brief explanation of the overall and pairwise results.
- A 150-word procurement memo with a practical threshold, assumptions and a proposed follow-up.
- If a minimum strength threshold is not supplied, request one; do not invent an engineering requirement.